Snowflake Project – Data Engineering & SQL Solution

Date	28 October 2026
Team ID	EN25CS3010655
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	5 Marks

This section evaluates the quality and completeness of the implemented Snowflake solution. The submission should demonstrate that the data has been prepared, cleaned, transformed, validated, and made ready for downstream analysis..

Instructions

1. Ensure SQL scripts are well-structured, clearly organized, and follow appropriate SQL practices.
2. The solution should address the project requirements through the required data preparation and transformation steps.
3. Include all SQL scripts, table/view definitions, and setup information needed to reproduce the solution.
4. Provide evidence that the final Snowflake data is accurate, consistent, and ready for analysis.

Solution Summary

Field	Details
Project / Repository Link	https://github.com/shwetasinghchauhan1110/Health360-Comprehensive-Healthcare-Risk-Disease-Analytics.git
Data Source(s) Link	Smart Healthcare Dataset (~5,000 anonymized patient records with clinical vitals and lifestyle indicators)
Database	HEALTH360_DB (Schemas: RAW, STAGING, CURATED, MART)
Warehouse	HEALTH360_WH (Virtual Warehouse: MEDIUM size, auto-suspend & auto-resume enabled)
Tables / Views Created	1. HEALTH360_RAW.PATIENT_HEALTH_DATA 2. HEALTH360_STAGING.STG_PATIENT_HEALTH_CLEAN 3. HEALTH360_CURATED.PATIENT_RISK_MA

	RT 4. HEALTH360_MART.VW_DISEASE_ANALYTICS_SUMMARY
Key Transformations / Calculations	Median value imputation for missing vitals, string standardization & trimming, age-group bucketing (18-35, 36-50, 51-65, 65+), physiological range bounding, and composite disease risk scoring calculations.
KPIs created	1. Diabetes Rate: 34.9% 2. Average BMI: 24.97 3. Average Blood Pressure: 134.4 mmHg 4. Total Population Volume: 5,000 Patients
Key Insights	1. Disease prevalence is highest in the 65+ age demographic across Diabetes, Heart Disease, and Stroke. 2. Average Diabetes rate stands at 34.9%, Heart Disease at 30.0%, and Stroke at 24.9% across the 5,000 patient cohort. 3. Patient vitals (Blood Pressure avg 134.4 mmHg, Glucose, BMI avg 24.97) display clear distributions by gender and exercise levels.

Snowflake Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Required data is successfully loaded and accessible in Snowflake	Yes
2	Data is cleaned, transformed, and prepared appropriately	Yes
3	Required tables, views, and data relationships are correctly implemented	Yes
4	Required SQL logic, calculations, and KPIs are correctly implemented	Yes
5	Data and solution outputs are validated against the project requirements	Yes
6	Solution is properly organized, documented, and functional	Yes